Acceleration

In everyday language, **acceleration is a change in speed.** In science, acceleration is the rate of change of velocity, and like displacement and velocity, it need to be stated with a direction.

To simplify our ideas of acceleration, we will only consider acceleration in a straigt line. In this case, acceleration can be calculated uysing the formula:

Acceleration = Change in speed = Final speed - Initial Speed = *v - u = a*
Time Time *t*

where *v* is the final speed, *u* is the initial speed, *t* is time and *a* is the acceleration.

A **positive acceleration** means something is **speeding up** and a **negative acceleration** means something is **slowing down.**

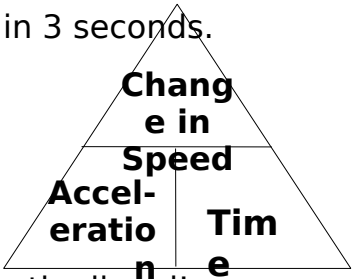
The **standard units** for acceleration are **m/s²**, although other units can be used.

Worked example 1 A car, which is at rest, speeds up to 15m/s in 5 seconds. Calculate its average acceleration in m/s^2.

$$a = \frac{v - u}{t} = \frac{15 - 0}{5} = 3 \text{ m/s}^2$$

Worked example 2 A cyclist decelerates from 12m/s to 6 m/s in 3 seconds. Calculate its deceleration in m/s^2.

$$a = \frac{v - u}{t} = \frac{6 - 12}{3} = \frac{-6}{3} = -2 \text{ m/s}^2$$



Calculating Acceleration

1. A lizard accelerates from 2m/s to 10m/s in 4 seconds. What is the lizard's acceleration?
2. A ball is dropped from the top of a building. After 2 seconds, its velocity is measured to be 19.6 m/s. Calculate the acceleration of the dropped ball.
3. As a bus leaves a stop, it speeds up from 0m/s to 9m/s in 5 seconds. What is the acceleration of the bus?
4. During a race, a sprinter increases his speed from 5 m/s to 7.5 m/s over a period of 1.25 seconds. What is the sprinter's acceleration during this period?
5. A car advertisement states that a certain car can accelerate from rest to 70km/h in 7 seconds.
   a. What is the car's speed in m/s?
   b. Find the car's acceleration.

6. A cyclist decelerates from 9m/s to 0m/s in 3 seconds. What is his deceleration?

7. A car manufacturer claims that its latest model can go from 0 to 90 km/h in 7.5 seconds.
    a. What is the car's final speed in m/s?
    b. What is the car's acceleration in m/s^2?



8. A roller coaster car rapidly picks up speed as it rolls down a hill. When it starts down the slope, its speed is 4m/s. 3 seconds later, at the bottom of the slope, its speed is 22m/s. What is its acceleration?

9. You are travelling in a car that is moving at a speed of 20m/s. You slam on your brakes and slow to 5m/s in 2 seconds. Calculate your deceleration.

10. A fighter jet landing on an aircraft carrier's flight deck that has a length of 300.0 meters must reduce its speed from 42.5 m/s to 0.0m/s in 2.0 seconds. What is the jet's deceleration?

Calculating Speed

1. A cyclist accelerates at 2m/s^2 during a 5 second interval. What is the change in speed of the cyclist?

2. A car with an initial velocity of 10 m/s, accelerates at a rate of 50 m/s^2 for 3 seconds.
    a. What is its change in speed?
    b. What will its final speed be?

3. Josh rolled a bowling ball down a lane in 2.5 s. The ball travelled at a constant acceleration of 1.8 m/ s^2 down the lane and was travelling at a speed of 7.6 m/s by the time it reached the pins at the end of the lane.
    a. What is the change in speed of the bowling ball?
    b. How fast was the ball going when it left Tim's hand?

4. The lifts in the Landmark Tower in Japan are among the fastest in the world. They accelerate upward at 3.125 m/s^2 for 4.0 seconds to reach their final speed.
    a. What is the change in speed of the lift?
    b. If these lifts start at rest, what is their final speed?



5. A freight train travelling with a speed of 18.0 m/s begins to accelerate at 0.4m/s^2. What is the train's speed after 23 seconds?

Calculating Time

1. Falling objects drop with an average acceleration of 9.8 m/ s^2. If an object falls from a tall building, how long will it take before it reaches a speed of 49 m/s?

2. How long would it take a car accelerating at 1.4m/s^2 to reach a speed of 21 m/s if it started from a standstill?

3. A kid sleds down a hill with an acceleration of 2.82 m/s^2. If her final speed is 15.5 m/s and she starts from rest, how long does it take to get to the bottom?

4. A helicopter rises from the ground with an acceleration of 2 m/s^2 until it reaches a speed of 17 m/s. How long would this take?

5. A remote control airplane is sitting on the runway and the operator revs the engine to full thrust then releases the brake. The plane accelerates down the runway at 1.6 m/s^2 and reaches its take off speed at 23m/s. How long did it take the plane to take off from the moment the brake was released?

Change in Speed

Accel-eration

Time